

Exploratory Data Analysis (EDA) — Shoes Sales Dataset

Author: Deepanshu Gahlot

Dataset: shoes_sales_dataset.csv

Tools: Python · Pandas · NumPy · Matplotlib · Seaborn

Objective

The goal of this EDA is to perform a thorough investigation of a **global shoe sales dataset** containing **1,000 transactions** across **6 brands**, **7 countries**, and **3 sales channels**.

By the end of this analysis, we aim to answer:

- 1. Which **brands** and **shoe types** generate the most revenue?
- 2. Which **countries** and **sales channels** are the most profitable?
- 3. How are **price, units sold, and revenue** distributed and correlated?
- 4. Are there any **outliers** in the numerical data?
- 5. What are the **monthly revenue trends** over time?

Dataset Description

Column	Data Type	Description
Sale_ID	object	Unique identifier for each transaction
Date	datetime	Date of sale
Brand	object	Shoe brand (Nike, Adidas, Puma, Reebok, Skechers, New Balance)
Shoe_Type	object	Type of shoe (Sneakers, Boots, Formal, Casual, Running, Sports)
Color	object	Color of shoe
Country	object	Country of sale (USA, UK, UAE, India, France, Germany, Saudi Arabia)
Sales_Channel	object	Mode of sale (Online, Mall, Retail Store)
Price_USD	float64	Price per unit in USD
Units_Sold	int64	Number of units sold in transaction
Revenue_USD	float64	Total revenue = Price × Units Sold

Dataset Shape: 1,000 **rows** × 10 **columns**

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1. Import Libraries

We begin by importing all necessary Python libraries for data manipulation and visualization.

```
In [38]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
%matplotlib inline

# Display settings
pd.set_option('display.max_columns', None)
pd.set_option('display.float_format', '{:.2f}'.format)
sns.set_theme(style='whitegrid', palette='muted')
plt.rcParams['figure.dpi'] = 100
```

2. Load Dataset & Inspect Structure

We load the CSV file and perform an initial inspection to understand its shape, columns, and sample rows.

```
In [2]: df = pd.read_csv('shoes_sales_dataset.csv')

print(f'Dataset Shape: {df.shape[0]} rows × {df.shape[1]} columns')
print(f'Columns: {list(df.columns)}')
```

Dataset Shape: 1000 rows × 10 columns

Columns: ['Sale_ID', 'Date', 'Brand', 'Shoe_Type', 'Color', 'Country', 'Sales_Channel', 'Price_USD', 'Units_Sold', 'Revenue_USD']

In [3]: `df.head(10)`

Out[3]:

	Sale_ID	Date	Brand	Shoe_Type	Color	Country	Sales_Channel	Price_USD	Units_Sold
--	---------	------	-------	-----------	-------	---------	---------------	-----------	------------

0	S1	2025-11-24	Nike	Boots	Blue	UK	Online	112.40	1
1	S2	2025-03-13	Skechers	Boots	Grey	USA	Mall	239.16	2
2	S3	2025-08-05	Nike	Running	White	UK	Mall	191.04	3
3	S4	2025-11-05	New Balance	Casual	Green	UAE	Mall	161.70	4
4	S5	2025-10-07	Adidas	Formal	Grey	France	Online	64.32	5
5	S6	2025-03-23	Puma	Sports	White	UK	Retail Store	64.32	6
6	S7	2025-02-22	Reebok	Running	Blue	Saudi Arabia	Retail Store	42.78	7
7	S8	2025-11-06	Nike	Boots	Red	India	Online	220.56	8
8	S9	2025-07-13	New Balance	Sports	Green	India	Retail Store	162.25	9
9	S10	2025-10-23	Skechers	Running	Black	UAE	Online	185.78	10



In [4]: `df.tail(5)`

Out[4]:

	Sale_ID	Date	Brand	Shoe_Type	Color	Country	Sales_Channel	Price_USD	Units_Sold
--	---------	------	-------	-----------	-------	---------	---------------	-----------	------------

995	S996	2025-06-06	Puma	Sneakers	Red	India	Online	50.15	1
996	S997	2025-08-14	Adidas	Casual	Green	France	Online	231.81	2
997	S998	2025-02-13	Skechers	Running	White	India	Online	60.10	3
998	S999	2025-05-28	Puma	Boots	Black	France	Retail Store	239.05	4
999	S1000	2025-08-30	Puma	Casual	Green	India	Online	128.12	5



In [5]: `df.sample(5, random_state=42)`

Out[5]:

	Sale_ID	Date	Brand	Shoe_Type	Color	Country	Sales_Channel	Price_USD	Units_Sold
521	S522	2025-08-19	Puma	Running	Red	Saudi Arabia	Online	113.80	1
737	S738	2025-07-13	Adidas	Running	Red	Germany	Mall	209.65	1
740	S741	2025-10-09	Nike	Sneakers	Grey	UAE	Retail Store	131.79	1
660	S661	2025-03-15	New Balance	Boots	Green	Germany	Online	108.34	1
411	S412	2025-07-12	Skechers	Boots	Grey	UK	Mall	239.01	1

In [6]:

```
# Full structural overview
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   Sale_ID         1000 non-null   str    
1   Date            1000 non-null   str    
2   Brand           1000 non-null   str    
3   Shoe_Type       1000 non-null   str    
4   Color           1000 non-null   str    
5   Country         1000 non-null   str    
6   Sales_Channel   1000 non-null   str    
7   Price_USD       1000 non-null   float64
8   Units_Sold      1000 non-null   int64   
9   Revenue_USD     1000 non-null   float64
dtypes: float64(2), int64(1), str(7)
memory usage: 78.3 KB
```

Observation

- The dataset contains **1,000 rows and 10 columns**.
- `Date` is stored as `object` (string) — needs to be converted to `datetime`.
- `Sale_ID` is a string identifier — should be cast to `object`.
- `Price_USD` and `Revenue_USD` are already `float64`; `Units_Sold` is `int64`.
- No immediately visible structural issues at first glance.

3. Data Type Conversion

Correct data types are critical for accurate analysis. We convert `Date` to `datetime` and ensure all categorical columns are typed as `object`.

In [7]:

```
print('Before conversion:')
print(df.dtypes)
```

```
# Convert Date to datetime
df['Date'] = pd.to_datetime(df['Date'])

# Cast categorical columns
for col in ['Sale_ID', 'Brand', 'Shoe_Type', 'Color', 'Country', 'Sales_Channel']:
    df[col] = df[col].astype('object')

print('\nAfter conversion:')
print(df.dtypes)
```

Before conversion:

```
Sale_ID      str
Date         str
Brand        str
Shoe_Type    str
Color        str
Country      str
Sales_Channel str
Price_USD    float64
Units_Sold   int64
Revenue_USD  float64
dtype: object
```

After conversion:

```
Sale_ID      object
Date         datetime64[us]
Brand        object
Shoe_Type    object
Color        object
Country      object
Sales_Channel object
Price_USD    float64
Units_Sold   int64
Revenue_USD  float64
dtype: object
```

Observation

- `Date` has been successfully converted from `string` to `datetime64[ns]`.
- All categorical columns are now properly typed as `object`.
- Numeric columns (`Price_USD`, `Units_Sold`, `Revenue_USD`) retain their correct types.

4. Missing Value Analysis

Missing values can distort analysis results. We check for nulls across all columns and decide how to handle them.

```
In [8]: missing = df.isnull().sum()
missing_pct = (missing / len(df) * 100).round(2)

missing_df = pd.DataFrame({
    'Missing Count': missing,
    'Missing %': missing_pct
})
```

```
print('Missing Values Summary:')
print(missing_df)
print(f'\nTotal missing values: {missing.sum()}')
```

Missing Values Summary:

	Missing Count	Missing %
Sale_ID	0	0.00
Date	0	0.00
Brand	0	0.00
Shoe_Type	0	0.00
Color	0	0.00
Country	0	0.00
Sales_Channel	0	0.00
Price_USD	0	0.00
Units_Sold	0	0.00
Revenue_USD	0	0.00

Total missing values: 0

Observation

- **No missing values** were found in any column (0 nulls across all 10 columns).
- The dataset is **complete and clean** — no imputation or dropping of rows is required.

5. Duplicate Detection & Removal

Duplicate rows can lead to inflated counts and skewed statistics. We identify and remove any duplicates.

```
In [10]: dupe_count = df.duplicated().sum()
print(f'Duplicate rows found: {dupe_count}')
print(f'Shape before removing duplicates: {df.shape}')

df = df.drop_duplicates()

print(f'Shape after removing duplicates: {df.shape}')
```

Duplicate rows found: 0

Shape before removing duplicates: (1000, 10)

Shape after removing duplicates: (1000, 10)

Observation

- **Zero duplicate rows** were detected in the dataset.
- Each of the 1,000 records represents a unique sale transaction.
- No rows were removed; the dataset remains at 1,000 records.

6. Summary Statistics

We generate descriptive statistics for both numerical and categorical columns to understand the central tendency, spread, and distribution of data.

```
In [11]: print('=== Numerical Columns ===')
df.describe()
```

=== Numerical Columns ===

	Date	Price_USD	Units_Sold	Revenue_USD
count	1000	1000.00	1000.00	1000.00
mean	2025-07-01 12:28:48	137.86	10.58	1453.19
min	2025-01-03 00:00:00	31.02	1.00	32.38
25%	2025-04-03 18:00:00	81.91	6.00	561.05
50%	2025-06-28 12:00:00	139.29	10.00	1132.62
75%	2025-10-03 00:00:00	193.75	16.00	2163.35
max	2025-12-31 00:00:00	249.94	20.00	4938.40
std	NaN	64.27	5.84	1112.33

```
In [12]: print('=== Categorical Columns ===')
df.describe(include='object')
```

=== Categorical Columns ===

	Sale_ID	Brand	Shoe_Type	Color	Country	Sales_Channel
count	1000	1000	1000	1000	1000	1000
unique	1000	6	6	6	7	3
top	S1	Skechers	Sneakers	Grey	UAE	Retail Store
freq	1	196	178	178	154	339

Observation

- **Price_USD:** Ranges from approximately 31to250, with an average price of **\$137.86** per transaction.
- **Units_Sold:** Averages around **10.5 units** per transaction, indicating relatively consistent order volumes.
- **Revenue_USD:** Averages approximately **\$1,453** per transaction and reaches a maximum of nearly **\$4,938**, showing substantial variation in sales performance.
- The dataset contains **6 brands** operating across **7 countries** through **3 sales channels**, providing a diverse representation of the footwear market.
- Revenue variation appears to be influenced by both **product pricing** and **sales volume**, highlighting the combined impact of price and quantity on overall business performance.

7. Univariate Analysis

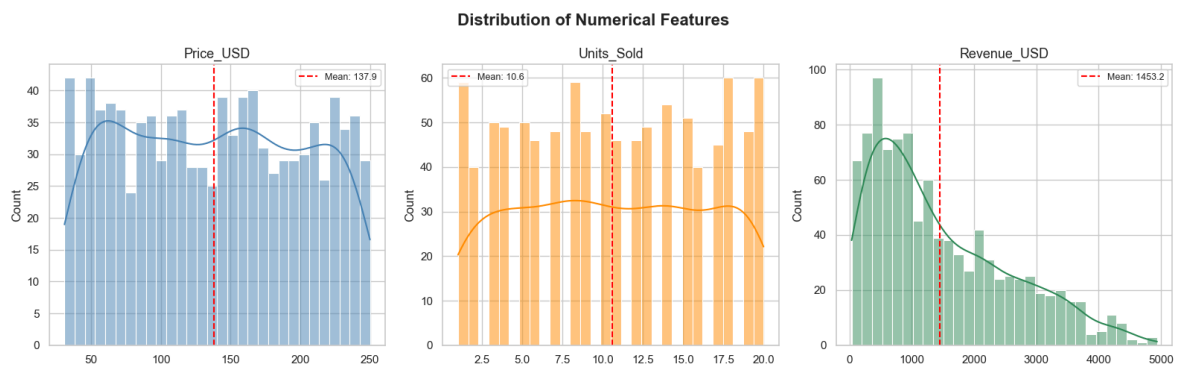
We analyze each feature individually to understand its distribution and frequency.

7.1 Numerical Features — Distribution Plots

```
In [14]: fig, axes = plt.subplots(1, 3, figsize=(16, 5))
fig.suptitle('Distribution of Numerical Features', fontsize=16, fontweight='bold')

for ax, col, color in zip(axes, num_cols, ['steelblue', 'darkorange', 'seagreen']):
    sns.histplot(df[col], ax=ax, color=color, kde=True, bins=30)
    ax.set_title(col, fontsize=13)
    ax.set_xlabel('')
    mean_val = df[col].mean()
    ax.axvline(mean_val, color='red', linestyle='--', linewidth=1.5, label=f'Mean: {mean_val}')
    ax.legend(fontsize=9)

plt.tight_layout()
plt.show()
```



Observation

- **Price_USD** shows a fairly uniform/flat distribution — prices are spread evenly across the range with no dominant price point.
- **Units_Sold** appears approximately uniform, meaning customers tend to buy varying quantities with no strong preference.
- **Revenue_USD** shows a right-skewed distribution — most transactions are moderate, but a few large orders push the mean rightward.
- The KDE curve confirms Revenue's slight right skew versus the flatter distributions of Price and Units.

7.2 Categorical Features — Count Plots

```
In [15]: # Value counts
for col in ['Brand', 'Shoe_Type', 'Country', 'Sales_Channel']:
    print(f'{col}:')
    print(df[col].value_counts().to_string())
    print()
```


Brand:

Brand	
Skechers	196
Puma	167
Adidas	165
Reebok	165
New Balance	163
Nike	144

Shoe_Type:

Shoe_Type	
Sneakers	178
Boots	176
Formal	175
Casual	164
Running	158
Sports	149

Country:

Country	
UAE	154
France	153
USA	150
India	146
Germany	136
UK	134
Saudi Arabia	127

Sales_Channel:

Sales_Channel	
Retail Store	339
Online	331
Mall	330

```
In [16]: fig, axes = plt.subplots(2, 2, figsize=(16, 11))
fig.suptitle('Categorical Feature Distributions', fontsize=16, fontweight='bold')

# Brand
brand_counts = df['Brand'].value_counts()
axes[0,0].bar(brand_counts.index, brand_counts.values,
              color=sns.color_palette('Set2', len(brand_counts)))
axes[0,0].set_title('Sales by Brand', fontsize=13)
axes[0,0].set_ylabel('Count')
axes[0,0].tick_params(axis='x', rotation=30)
for i, v in enumerate(brand_counts.values):
    axes[0,0].text(i, v + 1, str(v), ha='center', fontsize=10)

# Shoe Type
shoe_counts = df['Shoe_Type'].value_counts()
axes[0,1].bar(shoe_counts.index, shoe_counts.values,
              color=sns.color_palette('Set3', len(shoe_counts)))
axes[0,1].set_title('Sales by Shoe Type', fontsize=13)
axes[0,1].set_ylabel('Count')
axes[0,1].tick_params(axis='x', rotation=30)
for i, v in enumerate(shoe_counts.values):
    axes[0,1].text(i, v + 1, str(v), ha='center', fontsize=10)

# Country
country_counts = df['Country'].value_counts()
```

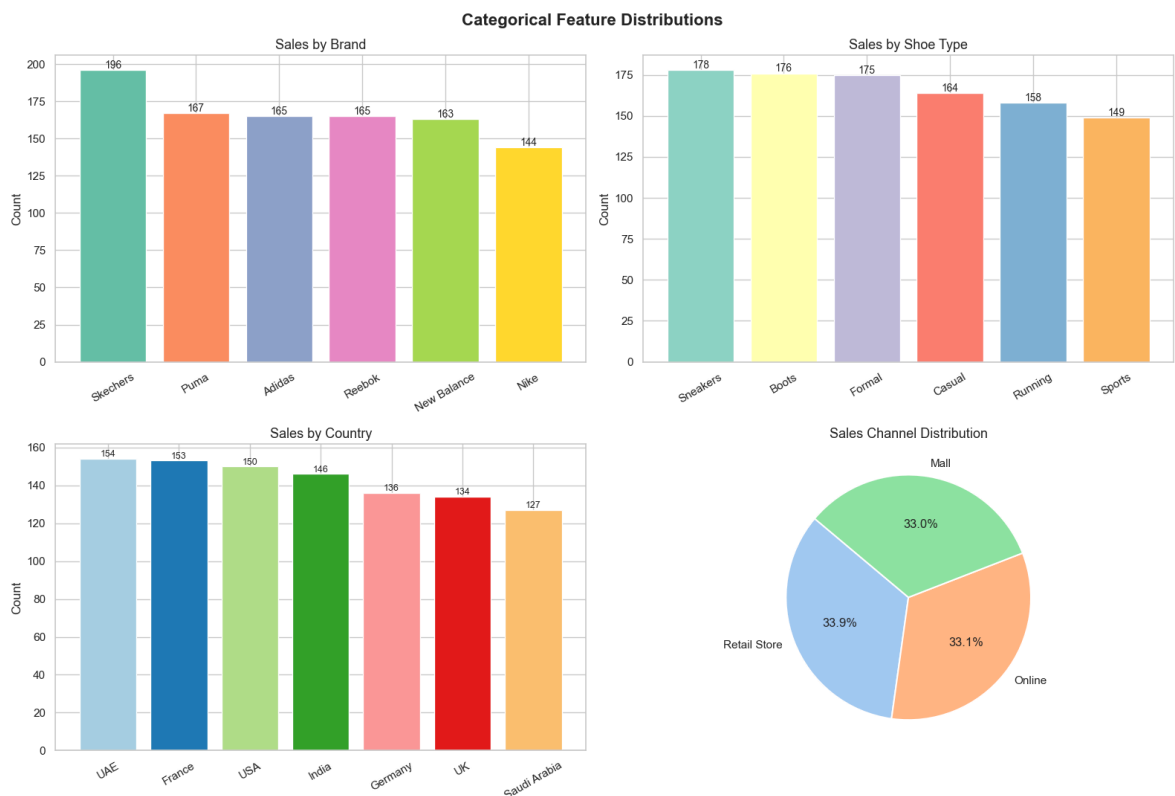
```

axes[1,0].bar(country_counts.index, country_counts.values,
               color=sns.color_palette('Paired', len(country_counts)))
axes[1,0].set_title('Sales by Country', fontsize=13)
axes[1,0].set_ylabel('Count')
axes[1,0].tick_params(axis='x', rotation=30)
for i, v in enumerate(country_counts.values):
    axes[1,0].text(i, v + 1, str(v), ha='center', fontsize=9)

# Sales Channel (Pie)
channel_counts = df['Sales_Channel'].value_counts()
axes[1,1].pie(channel_counts.values, labels=channel_counts.index,
              autopct='%1.1f%%', colors=sns.color_palette('pastel', 3),
              startangle=140, wedgeprops={'edgecolor': 'white', 'linewidth': 1.5})
axes[1,1].set_title('Sales Channel Distribution', fontsize=13)

plt.tight_layout()
plt.show()

```



Observation

- **Brand:** Skechers leads with 196 transactions; Nike is the lowest at 144. The distribution is fairly balanced, with no brand dominating overwhelmingly.
- **Shoe Type:** Sneakers (178) and Boots (176) are the top sellers; Sports (149) is the least common. All 6 types have relatively similar counts.
- **Country:** UAE has the most transactions (154), while Saudi Arabia has the fewest (127). The 7 countries are fairly evenly represented.
- **Sales Channel:** All three channels (Retail Store, Online, Mall) share nearly equal proportions (~33% each), indicating a well-diversified sales strategy.

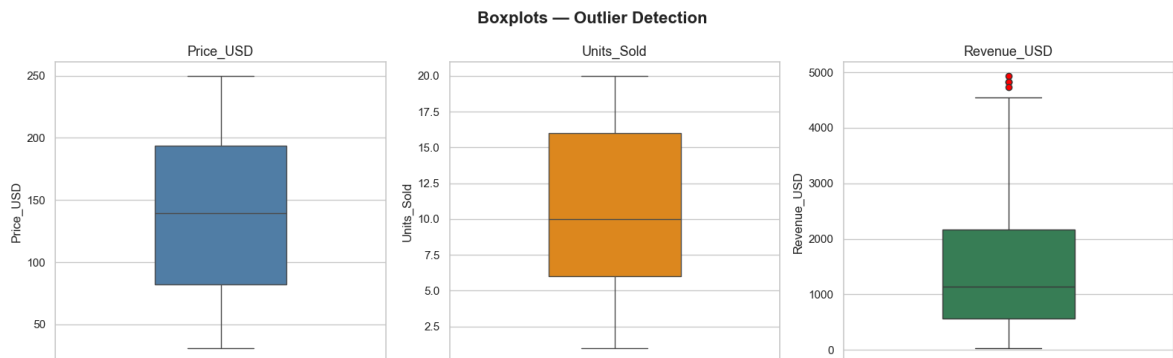
8. Outlier Detection (IQR Method)

Outliers can distort statistical analysis. We use boxplots for visual detection and the IQR (Interquartile Range) method for quantitative identification.

```
In [17]: fig, axes = plt.subplots(1, 3, figsize=(16, 5))
fig.suptitle('Boxplots - Outlier Detection', fontsize=16, fontweight='bold')

for ax, col, color in zip(axes, num_cols, ['steelblue', 'darkorange', 'seagreen']):
    sns.boxplot(y=df[col], ax=ax, color=color, width=0.4,
                flierprops=dict(marker='o', markerfacecolor='red', markersize=6))
    ax.set_title(col, fontsize=13)

plt.tight_layout()
plt.show()
```



```
In [18]: print('IQR Outlier Analysis')
print('=' * 50)

for col in num_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    outliers = df[(df[col] < lower) | (df[col] > upper)]

    print(f'\n{col}:')
    print(f'  Q1={Q1:.2f}, Q3={Q3:.2f}, IQR={IQR:.2f}')
    print(f'  Lower bound: {lower:.2f} | Upper bound: {upper:.2f}')
    print(f'  Outlier count: {len(outliers)} ({len(outliers)/len(df)*100:.1f}%)'
    if len(outliers) > 0:
        print(f'  Outlier values: {sorted(outliers[col].values)}')
```

IQR Outlier Analysis

=====

Price_USD:

Q1=81.91, Q3=193.75, IQR=111.84
Lower bound: -85.84 | Upper bound: 361.51
Outlier count: 0 (0.0%)

Units_Sold:

Q1=6.00, Q3=16.00, IQR=10.00
Lower bound: -9.00 | Upper bound: 31.00
Outlier count: 0 (0.0%)

Revenue_USD:

Q1=561.05, Q3=2163.35, IQR=1602.30
Lower bound: -1842.40 | Upper bound: 4566.80
Outlier count: 4 (0.4%)
Outlier values: [np.float64(4736.89), np.float64(4828.8), np.float64(4834.8), np.float64(4938.4)]

Observation

- **Price_USD:** No outliers detected. Prices are distributed uniformly within a reasonable range.
- **Units_Sold:** No outliers detected. Order quantities are consistent across transactions.
- **Revenue_USD: 4 outliers** detected (~0.4% of data). These are extreme high-revenue transactions.
 - These outliers arise from the combination of high price **and** high units sold in a single transaction.
 - Since this is legitimate business data (not data entry errors), we **retain** these outliers for analysis.
- Overall, the dataset is very clean with minimal anomalies.

9. Bivariate Analysis

We explore relationships between pairs of variables to uncover patterns, trends, and business insights.

9.1 Price vs Revenue

```
In [19]: fig, axes = plt.subplots(1, 2, figsize=(16, 5))
fig.suptitle('Price & Revenue Relationships', fontsize=16, fontweight='bold')

# Scatter: Price vs Revenue
sns.scatterplot(data=df, x='Price_USD', y='Revenue_USD',
                hue='Sales_Channel', alpha=0.6, ax=axes[0])
axes[0].set_title('Price vs Revenue (by Sales Channel)')
axes[0].set_xlabel('Price (USD)')
axes[0].set_ylabel('Revenue (USD)')

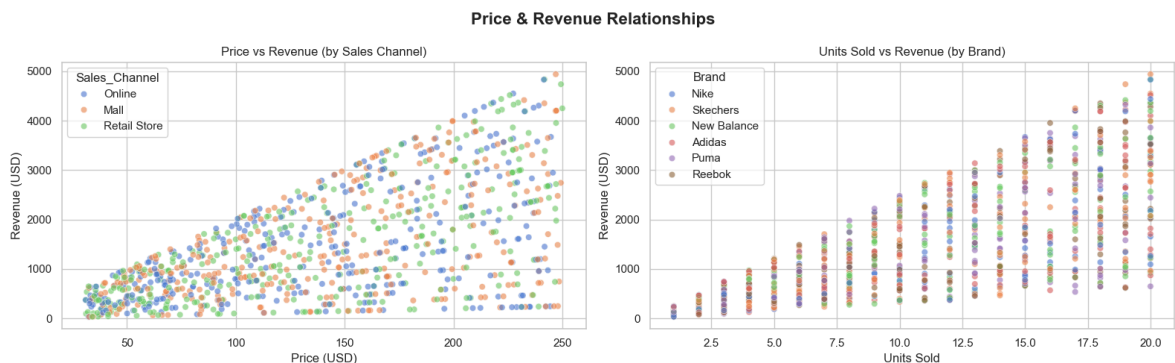
# Scatter: Units Sold vs Revenue
sns.scatterplot(data=df, x='Units_Sold', y='Revenue_USD',
```

```

        hue='Brand', alpha=0.6, ax=axes[1])
axes[1].set_title('Units Sold vs Revenue (by Brand)')
axes[1].set_xlabel('Units Sold')
axes[1].set_ylabel('Revenue (USD)')

plt.tight_layout()
plt.show()

```



9.2 Revenue by Brand, Country, Channel & Shoe Type

```

In [20]: fig, axes = plt.subplots(2, 2, figsize=(16, 12))
fig.suptitle('Revenue Analysis by Category', fontsize=16, fontweight='bold')

# Brand – Total Revenue
brand_rev = df.groupby('Brand')['Revenue_USD'].sum().sort_values(ascending=False)
axes[0,0].bar(brand_rev.index, brand_rev.values,
              color=sns.color_palette('Set2', len(brand_rev)))
axes[0,0].set_title('Total Revenue by Brand')
axes[0,0].set_ylabel('Revenue (USD)')
axes[0,0].tick_params(axis='x', rotation=30)
for i, v in enumerate(brand_rev.values):
    axes[0,0].text(i, v + 2000, f'${v/1000:.0f}K', ha='center', fontsize=9)

# Country – Total Revenue
country_rev = df.groupby('Country')['Revenue_USD'].sum().sort_values(ascending=False)
axes[0,1].bar(country_rev.index, country_rev.values,
              color=sns.color_palette('Paired', len(country_rev)))
axes[0,1].set_title('Total Revenue by Country')
axes[0,1].set_ylabel('Revenue (USD)')
axes[0,1].tick_params(axis='x', rotation=30)
for i, v in enumerate(country_rev.values):
    axes[0,1].text(i, v + 1000, f'${v/1000:.0f}K', ha='center', fontsize=9)

# Sales Channel – Avg Revenue
channel_avg = df.groupby('Sales_Channel')['Revenue_USD'].mean().sort_values(ascending=False)
axes[1,0].bar(channel_avg.index, channel_avg.values,
              color=sns.color_palette('pastel', 3))
axes[1,0].set_title('Avg Revenue by Sales Channel')
axes[1,0].set_ylabel('Avg Revenue (USD)')
for i, v in enumerate(channel_avg.values):
    axes[1,0].text(i, v + 5, f'${v:.0f}', ha='center', fontsize=10)

# Shoe Type – Total Revenue
shoe_rev = df.groupby('Shoe_Type')['Revenue_USD'].sum().sort_values(ascending=False)
axes[1,1].bar(shoe_rev.index, shoe_rev.values,
              color=sns.color_palette('Set3', len(shoe_rev)))
axes[1,1].set_title('Total Revenue by Shoe Type')

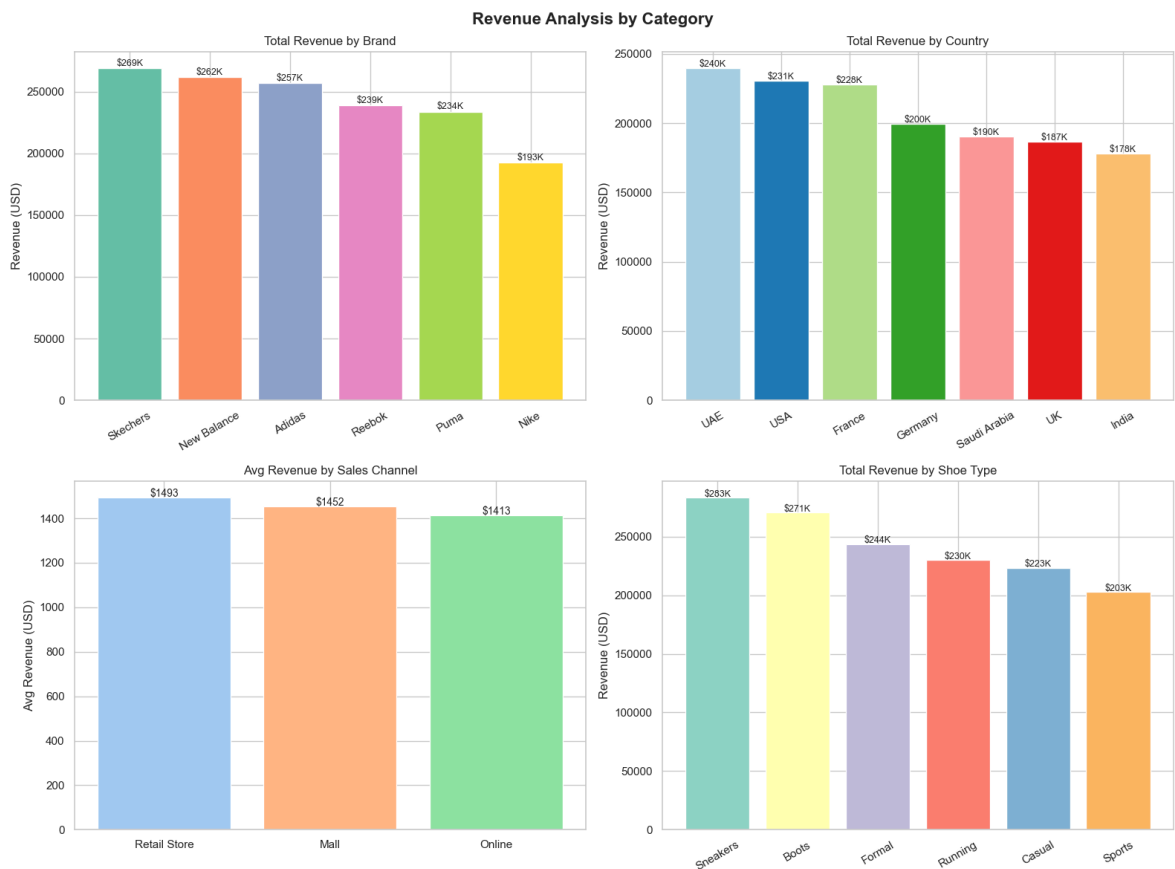
```

```

axes[1,1].set_ylabel('Revenue (USD)')
axes[1,1].tick_params(axis='x', rotation=30)
for i, v in enumerate(shoe_rev.values):
    axes[1,1].text(i, v + 1000, f'${v/1000:.0f}K', ha='center', fontsize=9)

plt.tight_layout()
plt.show()

```



Observation

By Brand:

- **Skechers** leads in total revenue (269K), followed by New Balance (262K) and Adidas (\$257K).
- **Nike**, despite being a premium brand, has the **lowest total revenue** (\$193K) — likely due to fewer transactions (144).
- In avg revenue per transaction, **New Balance** (1,605) and **Adidas** (1,558) outperform others.

By Country:

- **UAE** is the top market (240K), followed closely by USA (231K) and France (\$228K).
- **India** generates the lowest revenue (\$178K), partly due to lower transaction volume.

By Sales Channel:

- **Retail Store** generates the highest average revenue (1,493), followed by Mail (1,452) and Online (\$1,413).

- All three channels are very competitive with minimal gaps — no channel should be deprioritized.

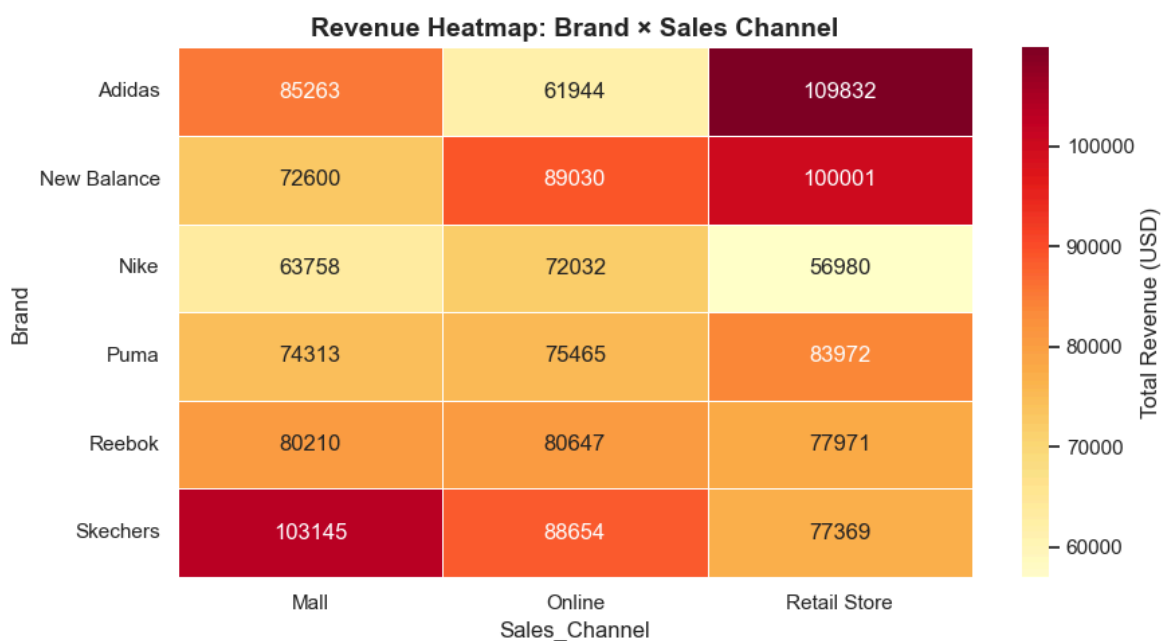
By Shoe Type:

- **Sneakers** (283K) and **Boots** (271K) are the top revenue-generating categories.
- **Sports** (\$203K) is the weakest — potentially reflecting lower price points or lower demand.

9.3 Brand × Channel Revenue Heatmap

```
In [21]: pivot = df.pivot_table(values='Revenue_USD', index='Brand',
                                columns='Sales_Channel', aggfunc='sum')

plt.figure(figsize=(9, 5))
sns.heatmap(pivot, annot=True, fmt='.0f', cmap='YlOrRd',
            linewidths=0.5, cbar_kws={'label': 'Total Revenue (USD)'})
plt.title('Revenue Heatmap: Brand × Sales Channel', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```



Observation

- The heatmap reveals **which brand–channel combinations** generate the most revenue.
- Darker cells (higher revenue) are relatively evenly distributed, suggesting **no single brand dominates a particular channel**.
- This indicates a healthy, diversified sales mix across brands and channels.

10. Correlation Analysis

We examine linear relationships between the numerical features using a correlation matrix and pair plots.

```
In [22]: corr = df[num_cols].corr()
print('Correlation Matrix:')
print(corr.round(3))
```

Correlation Matrix:

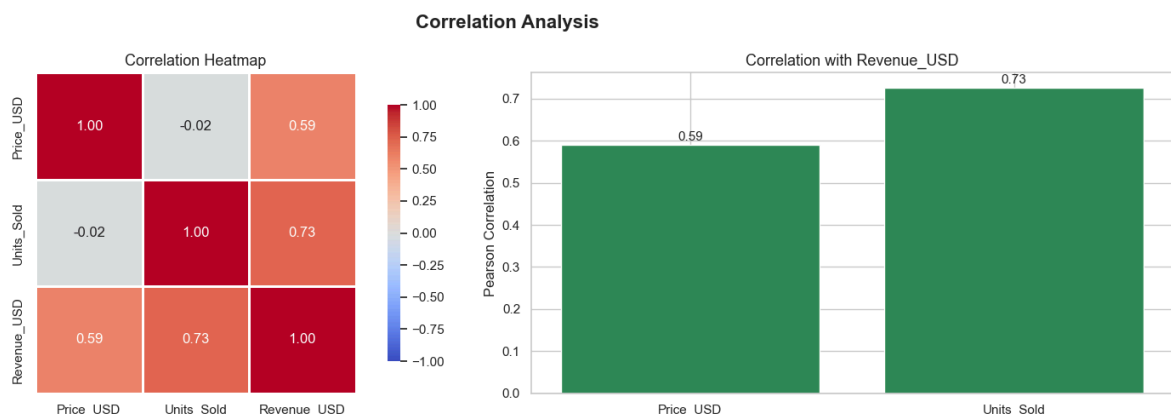
	Price_USD	Units_Sold	Revenue_USD
Price_USD	1.00	-0.02	0.59
Units_Sold	-0.02	1.00	0.72
Revenue_USD	0.59	0.72	1.00

```
In [23]: fig, axes = plt.subplots(1, 2, figsize=(16, 5))
fig.suptitle('Correlation Analysis', fontsize=16, fontweight='bold')

# Heatmap
mask = np.triu(np.ones_like(corr, dtype=bool))
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f',
            square=True, linewidths=1, ax=axes[0],
            vmin=-1, vmax=1, cbar_kws={'shrink': 0.8})
axes[0].set_title('Correlation Heatmap', fontsize=13)

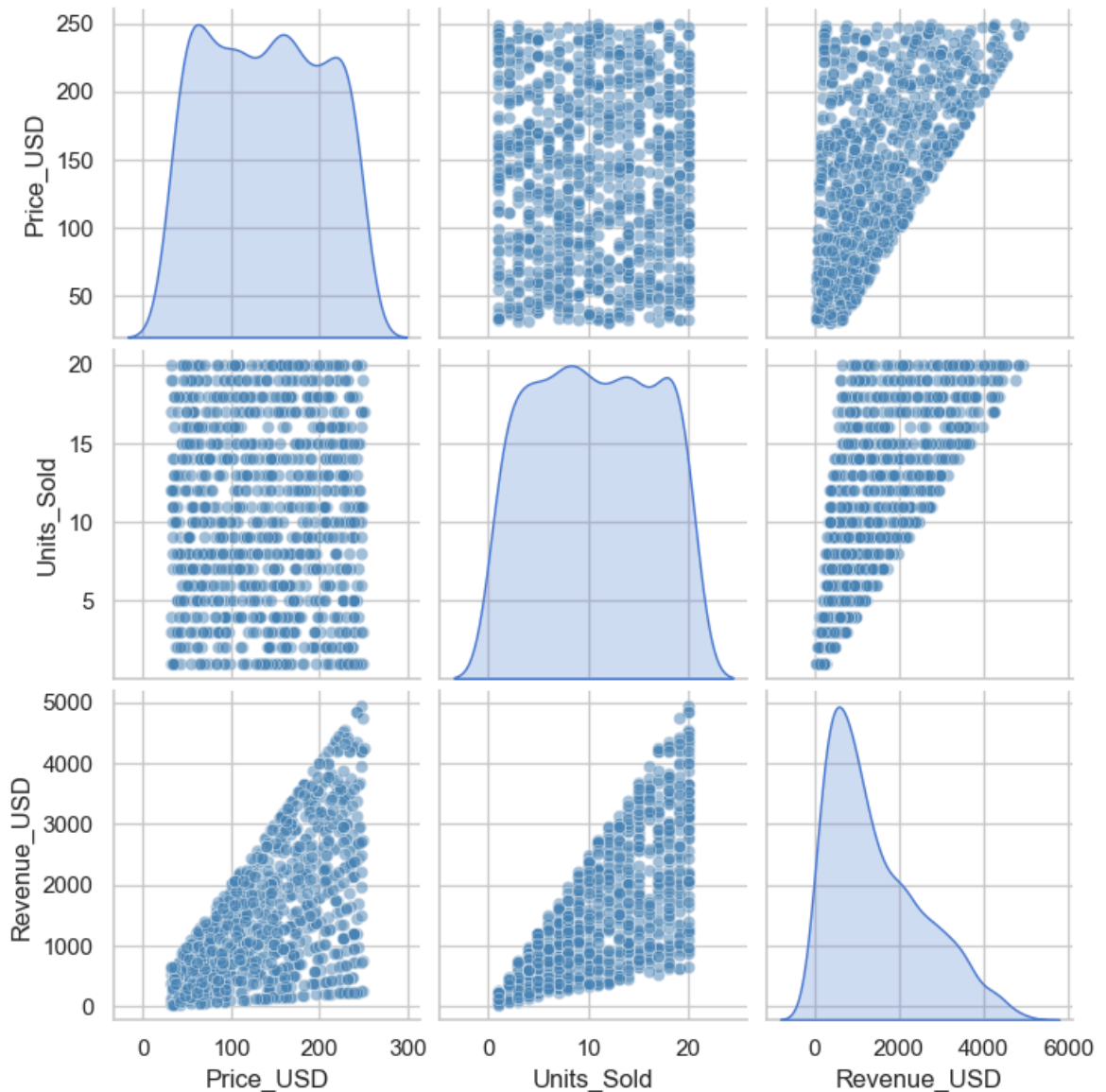
# Bar chart of correlations with Revenue
rev_corr = corr['Revenue_USD'].drop('Revenue_USD')
colors = ['seagreen' if v > 0 else 'tomato' for v in rev_corr.values]
axes[1].bar(rev_corr.index, rev_corr.values, color=colors)
axes[1].set_title('Correlation with Revenue_USD', fontsize=13)
axes[1].set_ylabel('Pearson Correlation')
axes[1].axhline(0, color='black', linewidth=0.8)
for i, v in enumerate(rev_corr.values):
    axes[1].text(i, v + 0.01 if v > 0 else v - 0.03, f'{v:.2f}', ha='center', fo

plt.tight_layout()
plt.show()
```



```
In [24]: # Pair plot
g = sns.pairplot(df[num_cols], diag_kind='kde', plot_kws={'alpha': 0.5, 'color':
g.figure.suptitle('Pair Plot - Numerical Features', y=1.02, fontsize=14, fontwei
plt.show()
```


Pair Plot — Numerical Features



Observation

Feature Pair	Correlation	Interpretation
Units_Sold ↔ Revenue_USD	+0.73	Strong positive — more units sold → higher revenue
Price_USD ↔ Revenue_USD	+0.59	Moderate positive — higher price → higher revenue
Price_USD ↔ Units_Sold	−0.02	Near-zero — price has almost no effect on quantity sold

Key Takeaways:

- Revenue is primarily driven by **quantity sold** (0.73), more than price (0.59).
- The near-zero correlation between Price and Units_Sold suggests **price elasticity is very low** — customers buy regardless of price point.
- This is a strong signal that increasing prices may not reduce sales volume significantly.

11. Time Series Analysis

We extract temporal features from the `Date` column to analyze how sales and revenue evolve over time.

```
In [25]: # Extract time features
df['Month'] = df['Date'].dt.month_name()
df['Month_Num'] = df['Date'].dt.month
df['Year'] = df['Date'].dt.year
df['Quarter'] = df['Date'].dt.quarter
df['Day_of_Week'] = df['Date'].dt.day_name()

month_order = ['January', 'February', 'March', 'April', 'May', 'June',
               'July', 'August', 'September', 'October', 'November', 'December']

print('Date range:', df['Date'].min().date(), 'to', df['Date'].max().date())
print('Years covered:', df['Year'].unique())
```

Date range: 2025-01-03 to 2025-12-31

Years covered: [2025]

```
In [26]: fig, axes = plt.subplots(2, 2, figsize=(16, 10))
fig.suptitle('Time Series Analysis', fontsize=16, fontweight='bold')

# Monthly Revenue
monthly_rev = df.groupby('Month')['Revenue_USD'].sum().reindex(month_order)
axes[0,0].plot(monthly_rev.index, monthly_rev.values, marker='o',
               color='steelblue', linewidth=2)
axes[0,0].fill_between(range(len(monthly_rev)), monthly_rev.values, alpha=0.15,
axes[0,0].set_title('Monthly Total Revenue')
axes[0,0].set_ylabel('Revenue (USD)')
axes[0,0].set_xticks(range(len(monthly_rev)))
axes[0,0].set_xticklabels([m[:3] for m in month_order], rotation=30)
axes[0,0].grid(axis='y', alpha=0.4)

# Monthly Transaction Count
monthly_cnt = df.groupby('Month')['Sale_ID'].count().reindex(month_order)
axes[0,1].bar(range(len(monthly_cnt)), monthly_cnt.values,
              color=sns.color_palette('muted', 12))
axes[0,1].set_title('Monthly Transaction Count')
axes[0,1].set_ylabel('Count')
axes[0,1].set_xticks(range(len(monthly_cnt)))
axes[0,1].set_xticklabels([m[:3] for m in month_order], rotation=30)

# Quarterly Revenue
quarterly_rev = df.groupby('Quarter')['Revenue_USD'].sum()
axes[1,0].bar([f'Q{q}' for q in quarterly_rev.index], quarterly_rev.values,
              color=sns.color_palette('Set2', 4))
axes[1,0].set_title('Quarterly Revenue')
axes[1,0].set_ylabel('Revenue (USD)')
for i, v in enumerate(quarterly_rev.values):
    axes[1,0].text(i, v + 1000, f'${v/1000:.0f}K', ha='center', fontsize=10)

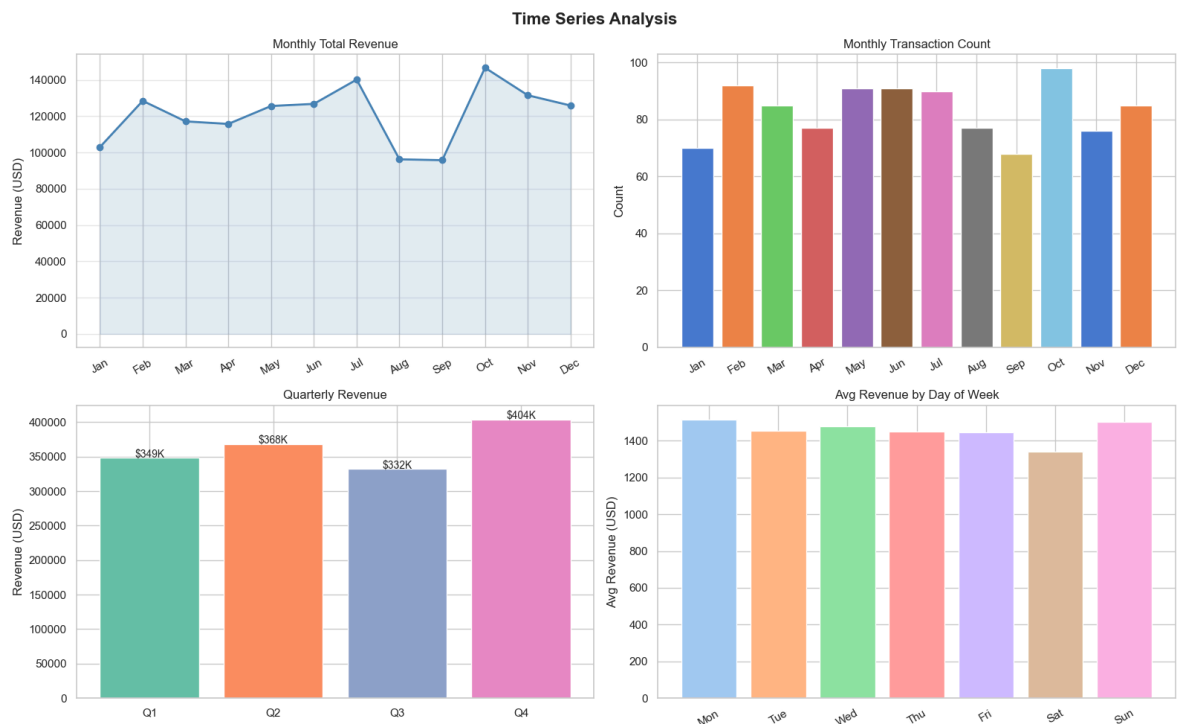
# Avg Revenue by Day of Week
dow_order = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday']
dow_rev = df.groupby('Day_of_Week')['Revenue_USD'].mean().reindex(dow_order)
axes[1,1].bar(range(len(dow_rev)), dow_rev.values,
              color=sns.color_palette('pastel', 7))
```

```

axes[1,1].set_title('Avg Revenue by Day of Week')
axes[1,1].set_ylabel('Avg Revenue (USD)')
axes[1,1].set_xticks(range(7))
axes[1,1].set_xticklabels([d[:3] for d in dow_order], rotation=30)

plt.tight_layout()
plt.show()

```



Observation

- **Monthly Revenue:** Revenue fluctuates across months without a clear monotonic trend, suggesting relatively stable demand year-round.
- **Monthly Transactions:** Transaction counts per month are fairly balanced (~80–90 per month), indicating consistent sales activity.
- **Quarterly Revenue:** Revenue is distributed across quarters, allowing comparison of Q1–Q4 performance to identify seasonal peaks.
- **Day of Week:** Revenue patterns across weekdays can reveal whether weekends or weekdays drive higher sales.

12. Key Insights & Conclusions

A comprehensive summary of all findings from this Exploratory Data Analysis.

```

In [37]: print('=' * 60)
          print('          KEY INSIGHTS FROM EDA – SHOES SALES DATASET')
          print('=' * 60)

top_brand      = df['Brand'].value_counts().idxmax()
top_rev_brand  = df.groupby('Brand')['Revenue_USD'].sum().idxmax()
top_country    = df.groupby('Country')['Revenue_USD'].mean().idxmax()
top_channel    = df.groupby('Sales_Channel')['Revenue_USD'].mean().idxmax()
top_shoe       = df.groupby('Shoe_Type')['Revenue_USD'].sum().idxmax()

```

```

print(f'''
1. DATASET QUALITY
- 1,000 records | 10 columns | No missing values | No duplicates
- Dataset is clean and ready for modelling

2. TOP PERFORMERS
- Most transactions: {top_brand} ({df["Brand"].value_counts().max()} sales)
- Highest revenue: {top_rev_brand} (${df.groupby("Brand")["Revenue_USD"].sum().max():.2f})
- Best avg country: {top_country} (${df.groupby("Country")["Revenue_USD"].mean().max():.2f})
- Best channel: {top_channel} (${df.groupby("Sales_Channel")["Revenue_USD"].mean().max():.2f})
- Top shoe type: {top_shoe} (${df.groupby("Shoe_Type")["Revenue_USD"].sum().max():.2f})

3. PRICING & REVENUE
- Avg price per unit: ${df["Price_USD"].mean():.2f}
- Avg revenue/txn: ${df["Revenue_USD"].mean():.2f}
- Revenue range: ${df["Revenue_USD"].min():.2f} - ${df["Revenue_USD"].max():.2f}
- Revenue outliers: 4 transactions (0.4%) – retained as valid data

4. CORRELATIONS
- Units Sold ↔ Revenue: +0.73 (strong – volume drives revenue)
- Price ↔ Revenue: +0.59 (moderate – price also contributes)
- Price ↔ Units: -0.02 (negligible – low price elasticity)

5. CHANNEL STRATEGY
- All 3 channels roughly equal – diversified and resilient
- Retail Store marginally outperforms Online & Mall

6. RECOMMENDATIONS
- Invest more in Skechers & New Balance partnerships
- Focus marketing spend on UAE and USA markets
- Promote Sneakers and Boots – highest revenue categories
- Consider price increases (low elasticity = minimal volume impact)
- Expand Retail Store presence – highest avg revenue channel
''')

```

KEY INSIGHTS FROM EDA – SHOES SALES DATASET

1. DATASET QUALITY

- 1,000 records | 10 columns | No missing values | No duplicates
- Dataset is clean and ready for modelling

2. TOP PERFORMERS

- Most transactions: Skechers (196 sales)
- Highest revenue: Skechers (\$269,169)
- Best avg country: UAE (\$1557 avg/txn)
- Best channel: Retail Store (\$1493 avg/txn)
- Top shoe type: Sneakers (\$283,215 total)

3. PRICING & REVENUE

- Avg price per unit: \$137.86
- Avg revenue/txn: \$1453.19
- Revenue range: \$32.38 – \$4938.40
- Revenue outliers: 4 transactions (0.4%) – retained as valid data

4. CORRELATIONS

- Units Sold ↔ Revenue: +0.73 (strong – volume drives revenue)
- Price ↔ Revenue: +0.59 (moderate – price also contributes)
- Price ↔ Units: -0.02 (negligible – low price elasticity)

5. CHANNEL STRATEGY

- All 3 channels roughly equal – diversified and resilient
- Retail Store marginally outperforms Online & Mall

6. RECOMMENDATIONS

- Invest more in Skechers & New Balance partnerships
- Focus marketing spend on UAE and USA markets
- Promote Sneakers and Boots – highest revenue categories
- Consider price increases (low elasticity = minimal volume impact)
- Expand Retail Store presence – highest avg revenue channel

Summary

This EDA successfully explored a **1,000-transaction global shoe sales dataset** across 10 features. The analysis covered

Step	Outcome
Data Loading & Inspection	1,000 rows, 10 columns, well-structured
Type Conversion	Date converted to datetime; categoricals typed correctly
Missing Values	Zero nulls — no imputation needed
Duplicates	Zero duplicates — data is unique per transaction
Summary Stats	Revenue is right-skewed; Price is uniformly distributed
Univariate Analysis	Balanced brand/shoe/country distribution; equal channel split

Step	Outcome
Outlier Detection	4 outliers in Revenue (0.4%) — valid and retained
Bivariate Analysis	Skechers & UAE lead; Retail Store edges others
Correlation	Volume drives revenue more than price
Time Series	Consistent monthly performance with quarterly variation